

Hawai'i's Electricity Future: Solar Reform, Enhanced Geothermal, and the JERA LNG Proposal

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All code, data, and results: github.com/mikejrob/oahu-electricity-future

Three decisions are on the table now

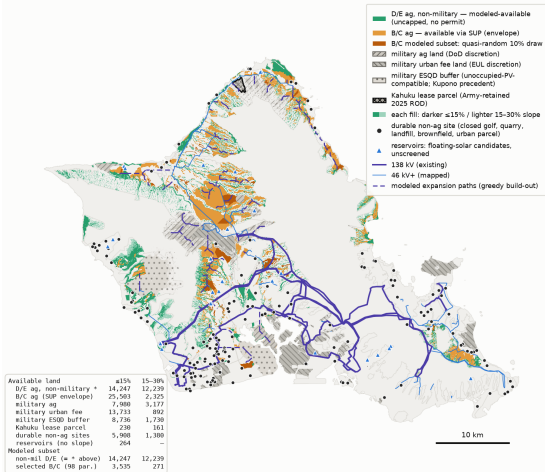
- **Waiau Repower** — six oil-fired turbines, approved March 2026
- **JERA proposal** — \$2B, 500 MW plant + LNG import terminal
- **Utility-scale solar** — procurement stalled; land rules bind

Open model, 2027–2050, 500+ solved scenarios. Everything public.

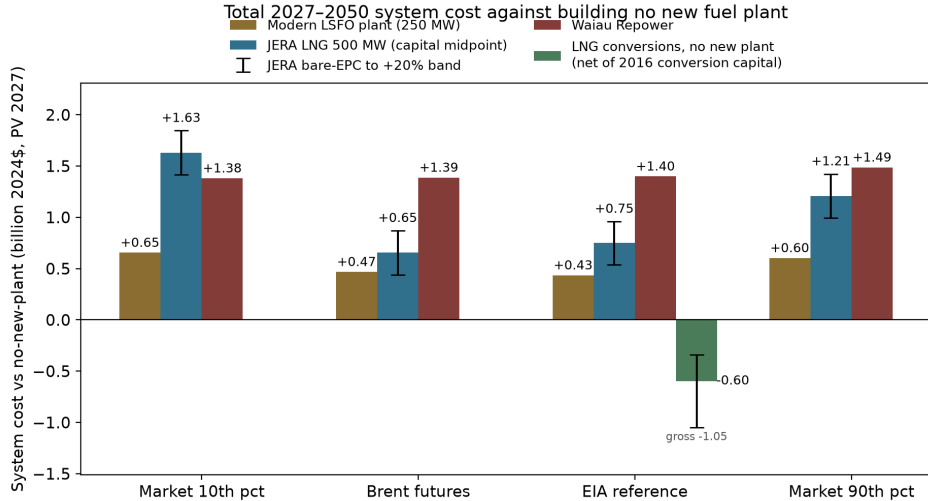
The land is there

Oahu: plausibly available solar land, the modeled subset, and transmission

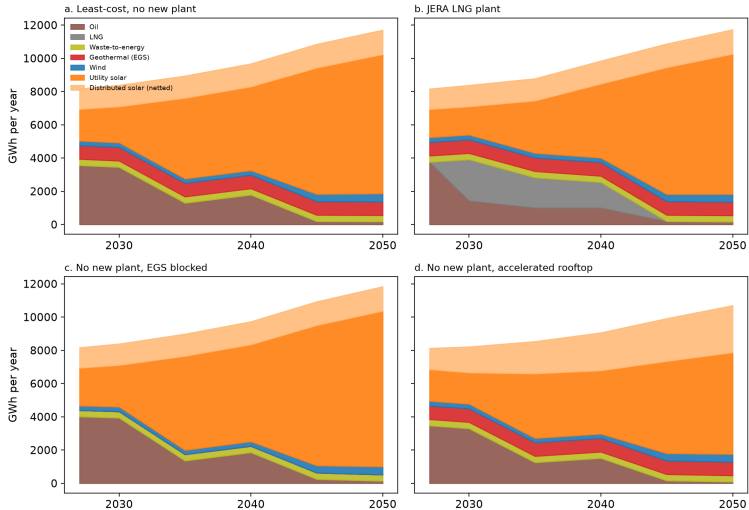
Each fill has two shades: darker = $\leq 15\%$ slope, lighter = the 15–30% increment (10 m DEM). The modeled subset is non-military D/E ag (green) plus a quasi-random 10% B/C draw; military land is shown by tenure category (available at DoD discretion, not in the modeled subset). No grid-distance filter: published near-grid figures are smaller (notes/available-land-map.mdi).



No proposed plant beats building no plant

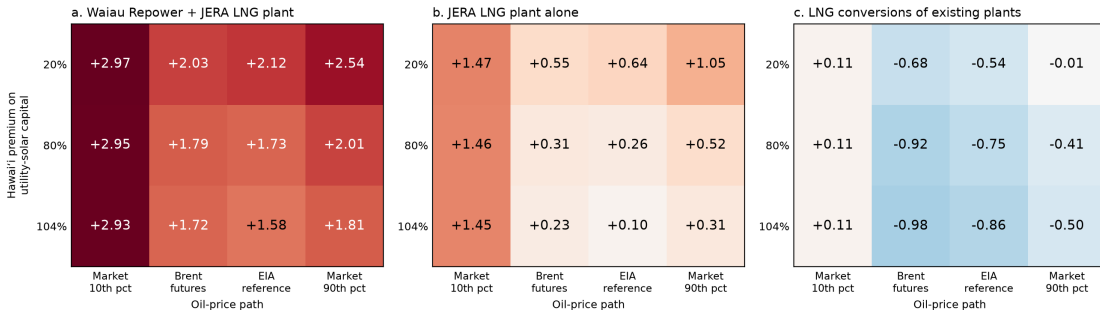


What the system builds instead



The verdict survives every oil and solar price

Cost of each thermal commitment against building no new plant (billion 2024\$; red costs more, blue saves)



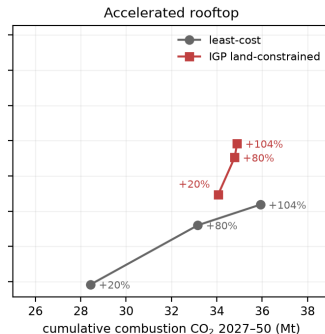
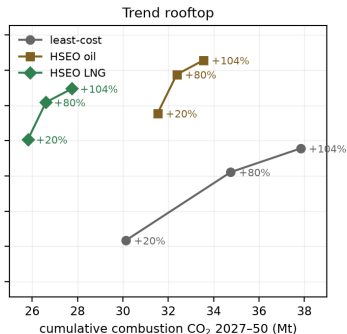
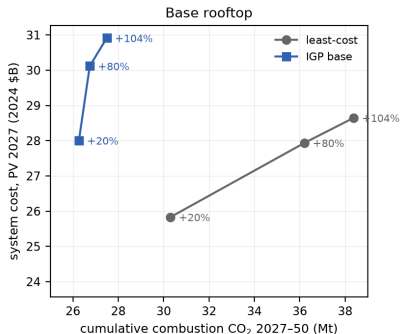
If LNG comes, existing plants beat new ones

LNG configuration (vs. no new plant)	Δ cost
New JERA plant (midpoint)	+\$0.7–1.6B
Convert Kalaeloa alone	–\$0.4B
+ Kahe 5 & 6, CIP turbine	–\$1.1B

And LNG's greenhouse advantage disappears above $\sim 0.9\%$ methane leakage — measured U.S. rates sit well above.

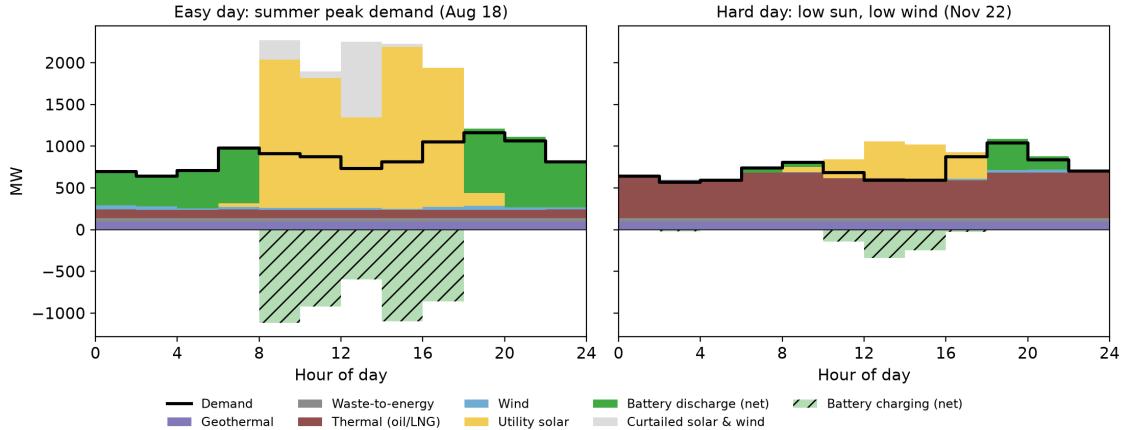
Every plan on file, priced on one ruler

Published plans vs the least-cost path as the solar premium rises: plans hold their mix (near-vertical); the optimum trades carbon for cost

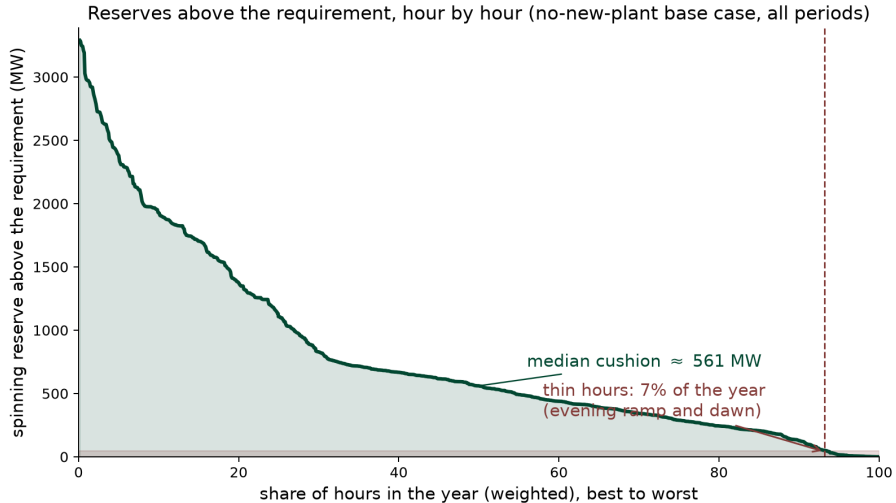


Reliability: the hard day is not the peak day

Hourly generation, no-new-plant base case, 2035



Resource adequacy: covered with room to spare



Four levers, no new technology required

- **Land:** siting rules exclude most flat, cleared, transmission-adjacent land
- **Procurement:** independent bid evaluation, standard contracts, published queues
- **Rooftops:** pay fair value for exported power
- **Wheeling:** settle at real marginal prices — Schedule RWS proposal in preparation

Takeaways

- **No proposed fuel plant pays for itself** — under any oil or solar price we can defend
- **Solar + storage + existing plants** is the robust least-cost core; geothermal sweetens it
- **The binding constraints are rules, not resources**

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Backup

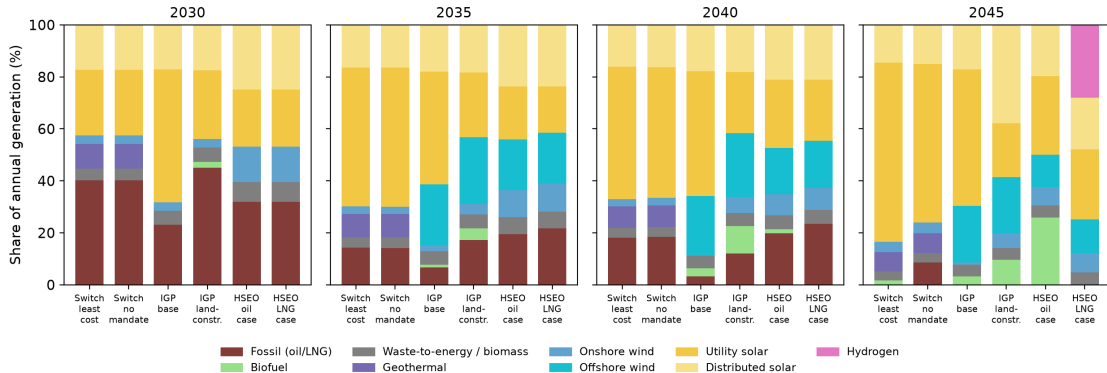
System cost by trajectory and oil price (Table ES.1)

	Market 10th	Brent futures	EIA ref.	Market 90th
No new fuel plant	21.2	24.3	25.8	27.3
Modern LSFO plant	+0.7	+0.5	+0.4	+0.6
JERA LNG (midpoint)	+1.6	+0.7	+0.8	+1.2
Waiau Repower	+1.4	+1.4	+1.4	+1.5
Waiau + JERA	+3.1	+2.1	+2.2	+2.7

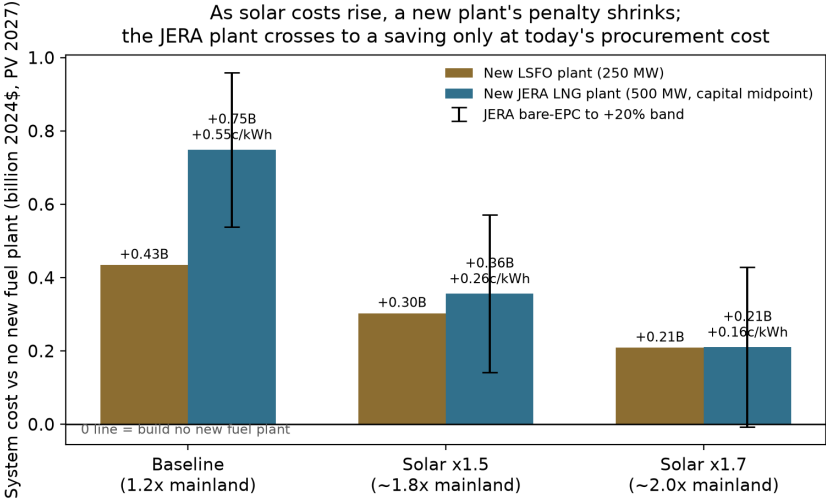
Billions of 2024\$, present value 2027–2050, 3% real.

Our least-cost paths vs. HECO's IGP and HSEO's study

Generation mix by plan: this report's solved paths vs HECO's IGP and HSEO's Alternative Fuels Study



Solar-cost sensitivity



Four oil futures, priced from markets

Four oil-price cases: EIA reference, futures central, and the market 10th/90th band
(real 2024\$, deflated on TIPS breakevens; σ beyond liquid tenors = 0.30, documented)

